

$E = n^\circ$ somadores
kernel $N \times N$

3 estágios

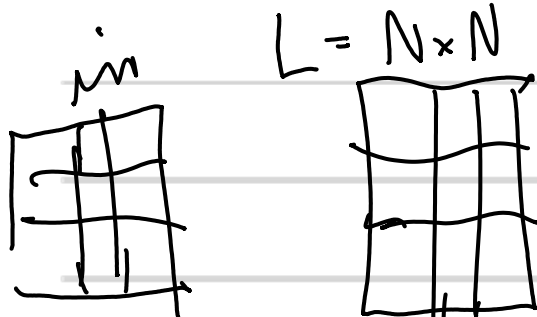
8 - 4 - 2 - 1

9 - 4 + 1 - 2 + 1 - 1 + 1 - 1

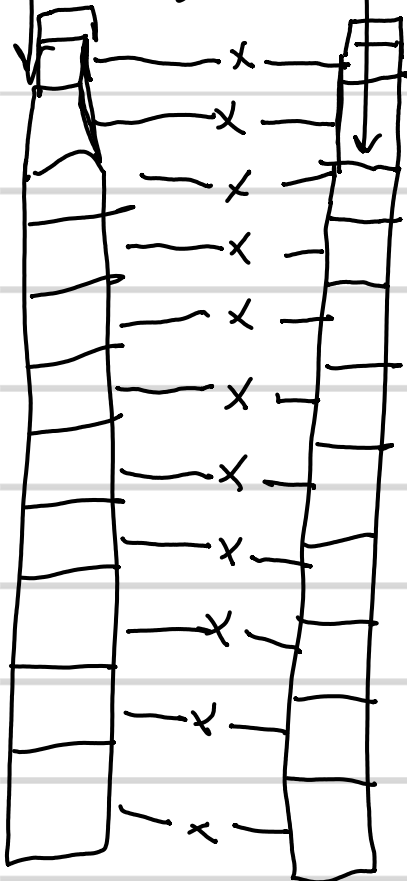
4 estágios

16 - 8 - 4 - 2 - 1

4 estágios



$$a_{11}k_{11} + a_{12}k_{12} + a_{13}k_{13} + \dots + a_{33}k_{33}$$



$$\frac{10}{2} = 5$$

$$\frac{5}{2} = 2R1$$

$$\frac{2}{2} = 1$$

$$R1 = 1$$

$$\frac{11}{2} = 5R1$$

$$\frac{5}{2} = 2R1$$

$$\frac{3}{2} = 1R1$$

$$R1 = 1$$

$$\frac{12}{2} = 6$$

$$\frac{6}{2} = 3$$

$$\frac{3}{2} = 1R1$$

$$R1 = 1$$

$$\frac{13}{2} = 6R1$$

$$\frac{6}{2} = 3$$

$$\frac{3}{2} = 1R1$$

$$\frac{2}{2} = 1$$

$$16: 8 + 4 + 2 + 1$$

$$8: 4 + 2 + 1$$

$$4: 2 + 1$$

$$9: 4 + 2 + 1 + 1$$

$$10: 5 + 2 + 1 + 1$$

$$11: 5 + 3 + 1 + 1$$

$$12: 6 + 3 + 1 + 1$$

$$13: 6 + 3 + 2 + 1$$

$$N^\circ \text{ estágios: } \lceil \log_2(L) \rceil$$

Nº nomadores de estágio = $\lfloor \frac{N^\circ \text{ elementos estágio anterior}}{2} \rfloor (+ 1 \text{ se } \sum \text{ resto} = 2)$

Se sobran resto, + 1 último estágio no final

0 1 2 3 4 5 6 7 8 9 10 11

$$\frac{11}{2} = 5 R_1$$

$$N = 11$$

$$\frac{N}{2} = 5 R_1$$

$$\frac{5}{2} = 2 R_1 \rightarrow 3$$

$$9 \gg 1 = 4 \quad 9 \gg$$

$$5 \gg 1 =$$

$$\frac{3}{2} = 1 R_1 \quad (x) + (x+1) \dots$$

$$R_1 = 1$$

for ($x = 0; x < \frac{N}{2}; x++$)

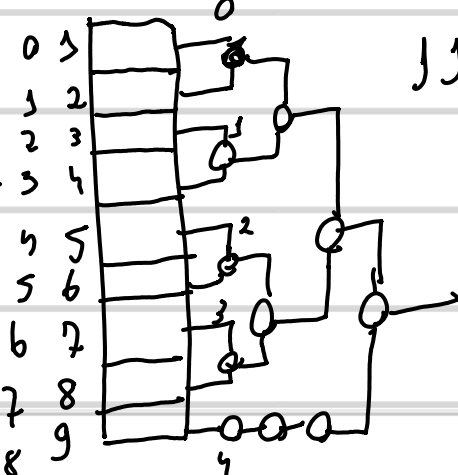
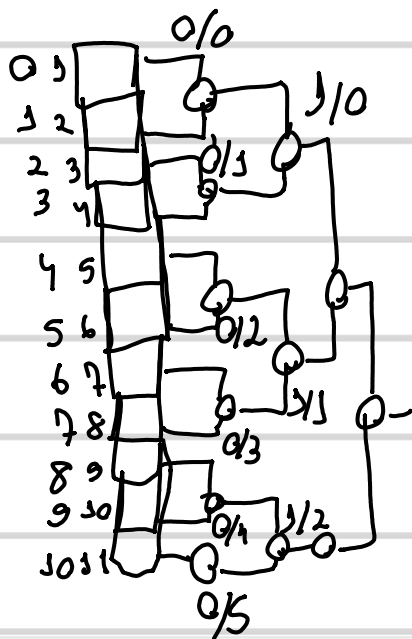
Soma[] = in[2x] + in[2x+1]

$$9 \gg 1 = 4 \gg 1 = 2 \gg 1 = 1$$

$$11 \gg 1 = 5 \gg 1 = 2$$

$$11 \gg 1 = 5 = 6 \gg 1 = 3 \gg 1$$

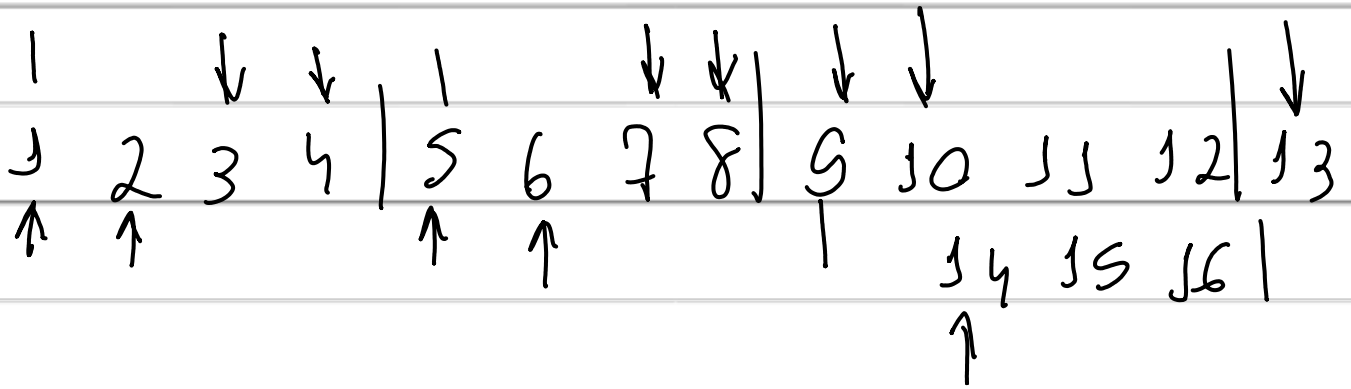
$$1 = 1 = 2 \gg 1 = 1$$



0	1	2	3	4	5
6	7	8	9	10	11
12	13	14	15	16	17
18	19	20	21	22	23
24	25	26	27	28	29
30	31	32	33	34	35

0	1	2	3	4	5
N	N+1	N+2	N+3	N+4	N+5
2N	2N+1	2N+2	2N+3	2N+4	2N+5
3N	3N+1	3N+2	3N+3	3N+4	3N+5
4N	4N+1	4N+2	4N+3	4N+4	4N+5
5N	5N+1	5N+2	5N+3	5N+4	5N+5

$i0j0$ $i0j2$ $i0j4$ $i2j0$ $i2j2$ $i2j4$ $i4j0$
 0 1 2 3 4 5 6



$\begin{array}{ccccccccc|cccccccc} & 1 & & & 1 & 1 & 1 & & & 1 & 1 & & & & \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 & 16 & 17 & 18 & 19 \\ & & \downarrow & \downarrow & & & & & \downarrow & \downarrow & & & & & & & & & \\ 20 & 21 & 22 & 23 & 24 & & 25 & 26 & 27 & 28 & 29 & 30 & 31 & 32 \\ & & & & & & & & & & & & & \\ 33 & 34 & 35 & 36 & & & & & & & & & & \end{array}$

$$0 \ 0 \ 0 \ 0 \quad 0$$

$$0 \ 0 \ 0 \ 1 \quad 1$$

$$0 \ 0 \ 1 \ 0 \quad 2$$

$$0 \ 0 \ 1 \ 1 \quad 3$$

$$0 \ 1 \ 0 \ 0 \quad 4$$

$$0 \ 1 \ 0 \ 1 \quad 5$$

$$0 \ 1 \ 1 \ 0 \quad 6$$

$$0 \ 1 \ 1 \ 1 \quad 7$$

$$1 \ 0 \ 0 \ 0 \quad -8$$

$$1 \ 0 \ 0 \ 1 \quad -7$$

$$1 \ 0 \ 1 \ 0 \quad -6$$

$$1 \ 0 \ 1 \ 1 \quad -5$$

$$1 \ 1 \ 0 \ 0 \quad -4$$

$$1 \ 1 \ 0 \ 1 \quad -3$$

$$1 \ 1 \ 1 \ 0 \quad -2$$

$$1 \ 1 \ 1 \ 1 \quad -1$$

0	1	2
3	4	5
6	7	8

i j pos
0 0 0

0 1 1

0 2 2

1 0 3

1 1 4

1 2 5

2 0 6

2 1 7

2 2 8

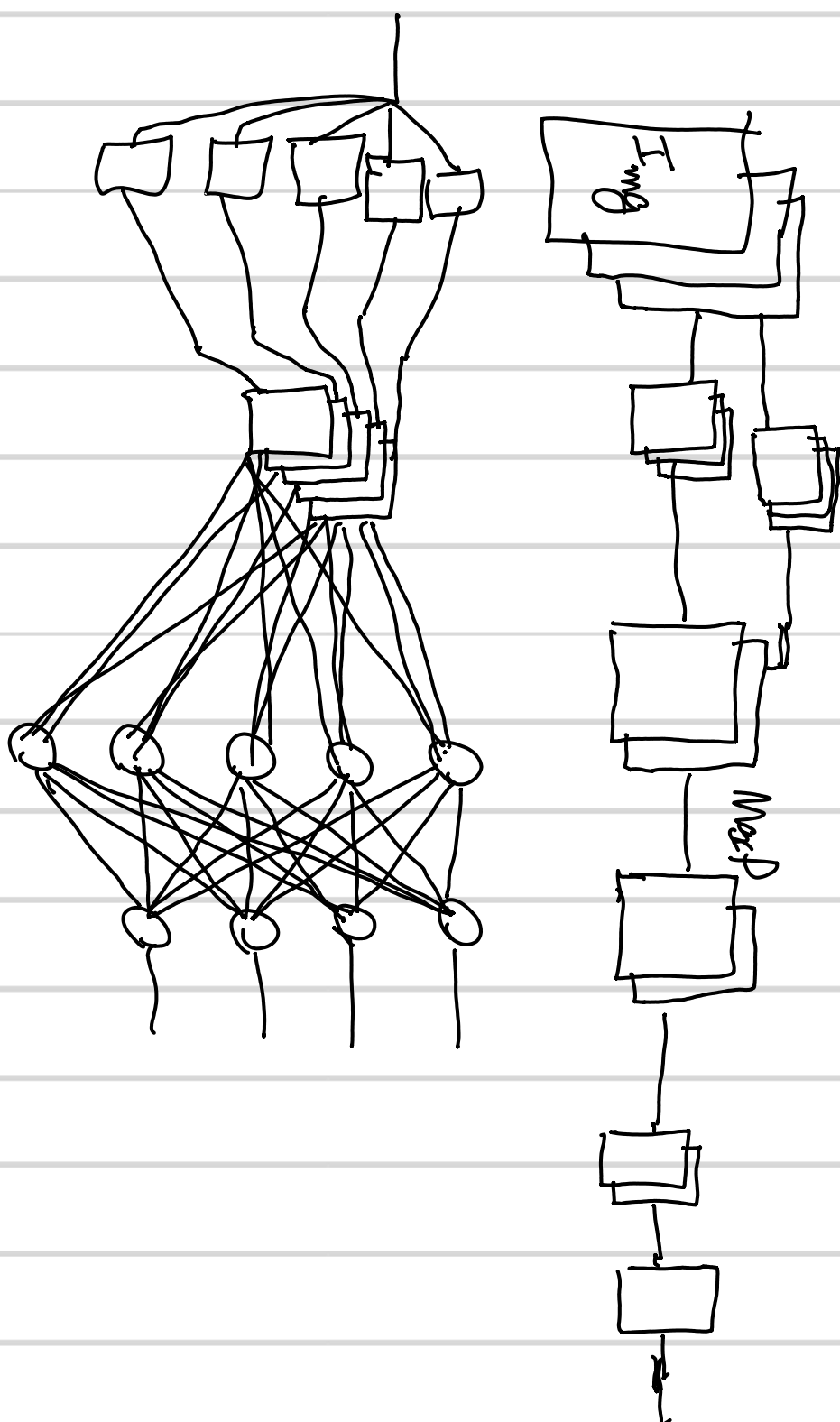
0	1	2	3	4
5	6	7	8	9
10	11	12	13	14
15	16	17	18	19
20	21	22	23	24



$$\begin{array}{r} 5-3+1 \\ 2 \end{array}$$

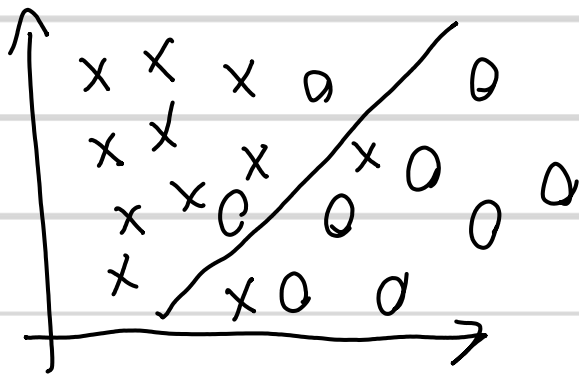
row 0 1 2 0 1 2 0 1 2
col 0 0 0 1 1 1 2 2 2

CNN

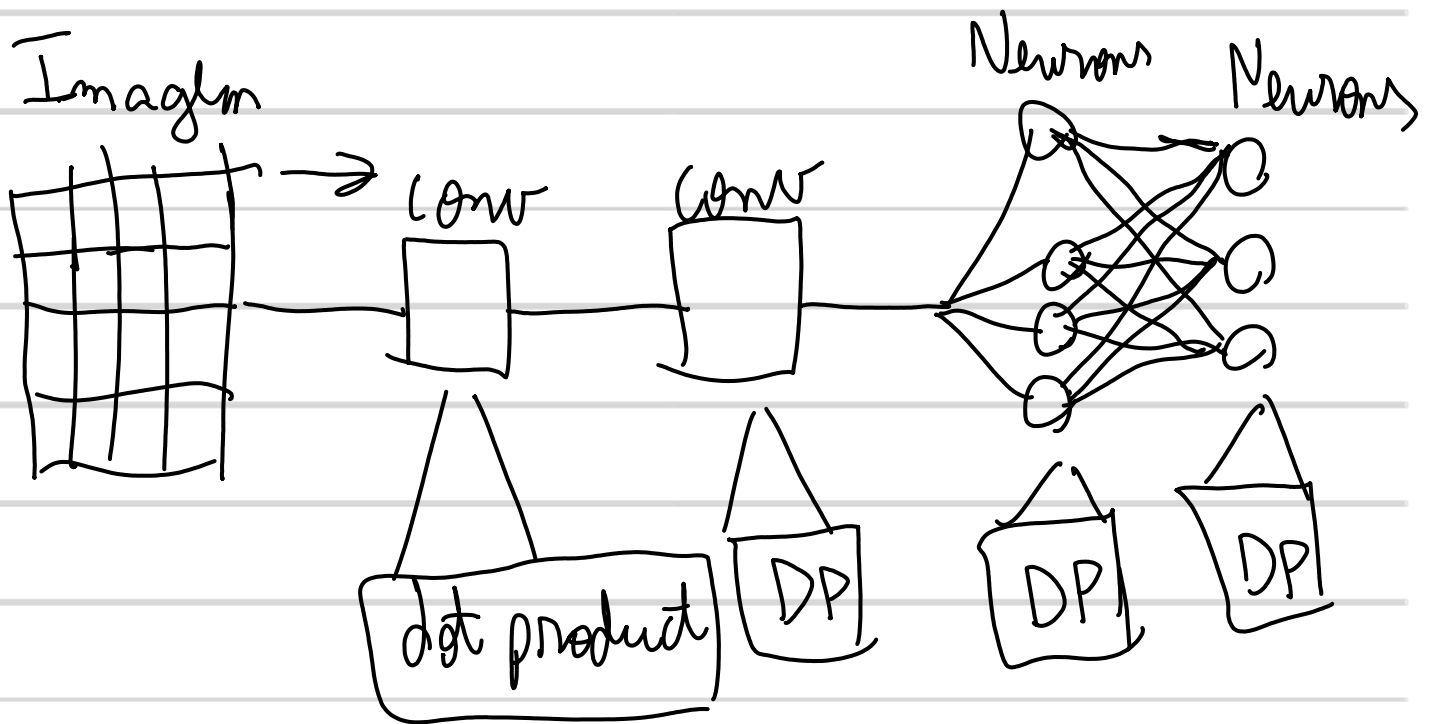


CNN diz: Número de neurônios, tamanho do kernel, tamanho do dado

X 0



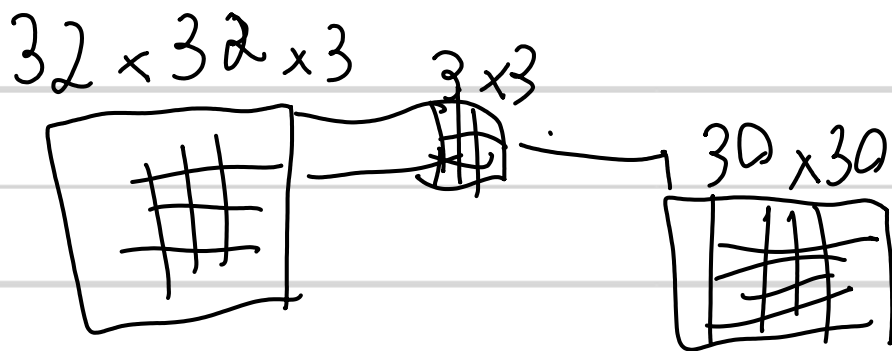
x x x x x 0 x x 0 x | 0 0 0 0 0



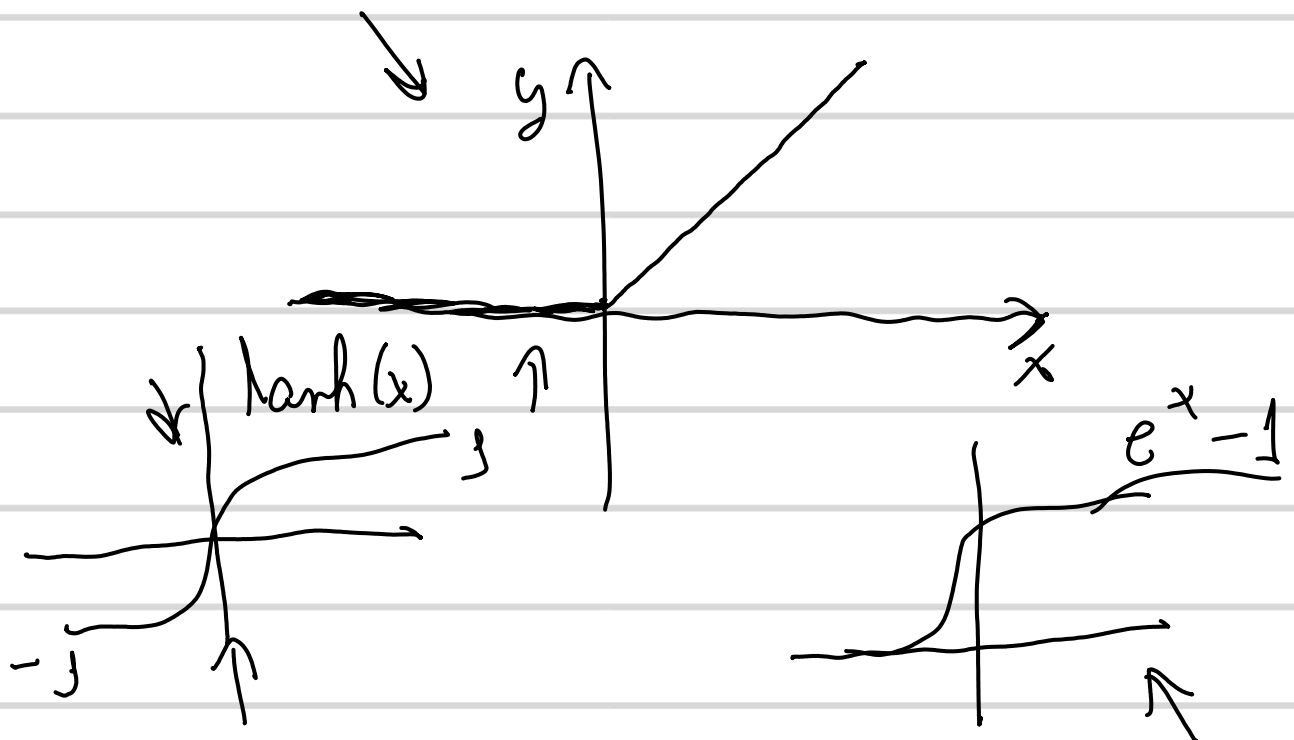
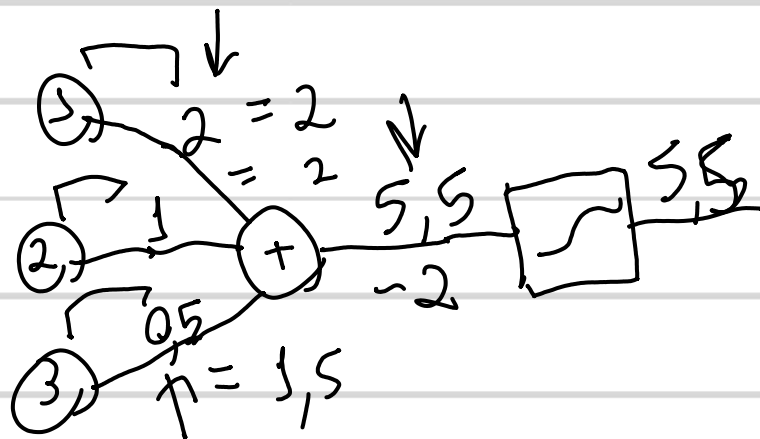
Quando chegar uma imagem válida, manda um start

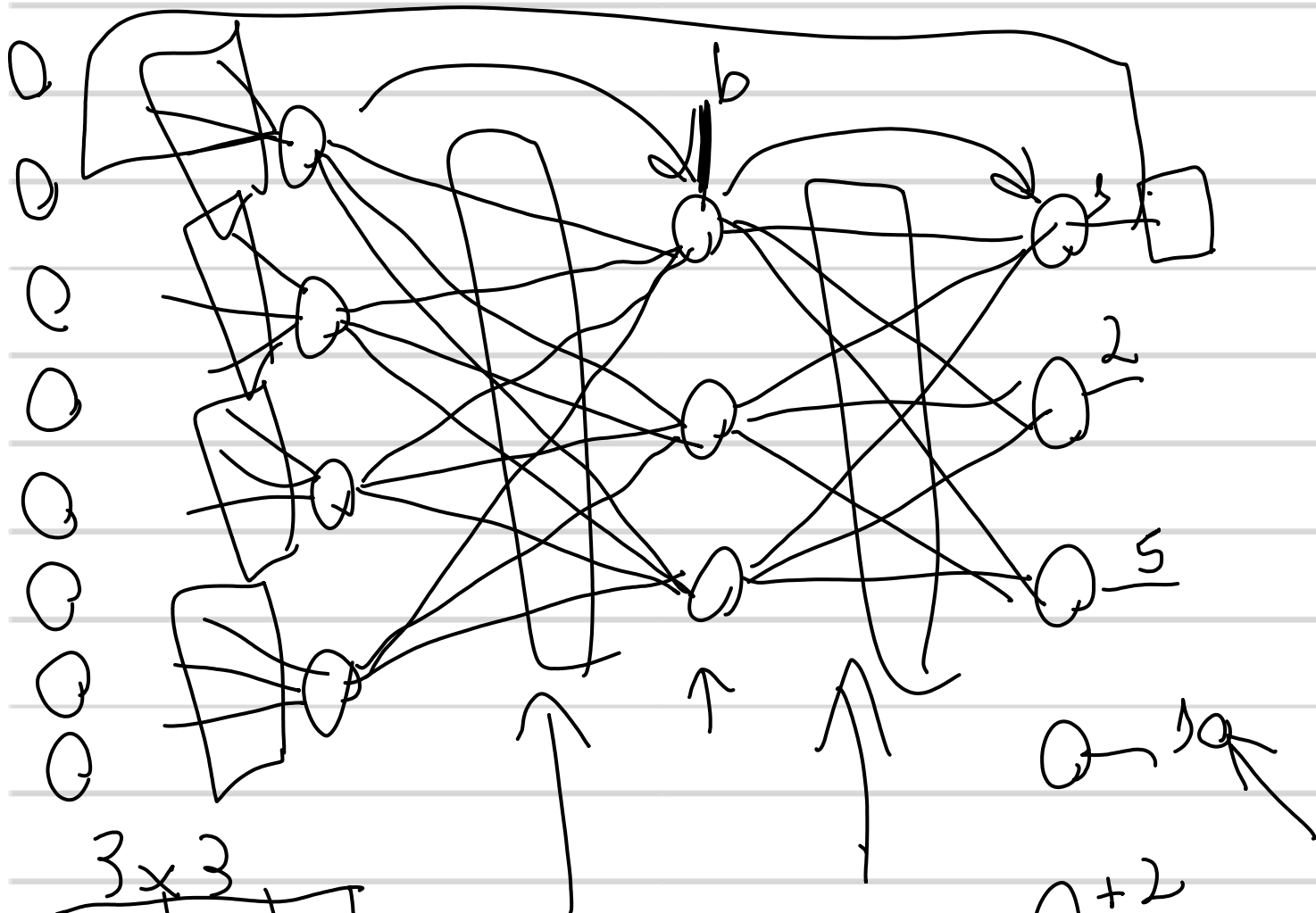
Conv1 recebe start, gera um valid_in, dot product trabalha até gerar um valid_out. Conv2 pega os valid_out de dot-product, monta seu ator convoluçao e gera seu valid_out.

Valid_out1 age como start p/ segunda camada.
Valid_out2 age como start para os neurônios



$$32 - 3 + 1 = 30$$





3x3

0	1	2
3	4	5
6	7	8

CIFAR-10

Q^{+1}
 Q^{+2}
 Q^7
 Q^8
 Q^8
 Q^3
 Q^1

0	1	2	3	4	5	6	7	8
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